comic book project

1 Methodology

This project uses two steps to make a short children comic book: text first, then images.

For text, we used Qwen2.5 7B Instruct in 4-bit to save memory. It generated 10 ordered panels in English. Each panel has a short scene description and short dialogue for two characters: Ziad and Omar.

For images, we used Stable Diffusion XL with SG161222 RealVisXL V4.0. We used DDIM, 30 steps, guidance 6.5, image size 512x512, and seed 200 for stable results.

To keep the same characters, we used two ideas: (1) Text locking: fixed age, face, hair, and clothes in every prompt, with the same comic style keywords. We also used a negative prompt to reduce errors like bad faces, bad hands, extra fingers, and watermarks. (2) Attention Bank: we used SpatialAttnProcessor2_0 with an attention bank. We set id_length = 4, so Panels 1–4 are identity reference frames. These frames are used to build identity memory, so they are not counted in the main evaluation. Later panels reuse the stored identity using sa32 = 0.85 and sa64 = 0.75.

2 Character Consistency Comparison

We compared three simple options: Method A: prompt locking + fixed seed + negative prompt. IP Adapter: uses a reference image. Our method: attention bank.

Table 1: Prompt Locking

Item	Value
Needs training	No
Needs ref image	No
Extra compute	Low
Identity stability	0.55
Best for	Simple scenes
Limitation	Face drift

Table 2: IP Adapter

Item	Value
Needs training	No
Needs ref image	Yes
Extra compute	Medium
Identity stability	0.85
Best for	Strong identity
Limitation	Style mismatch

Table 3: Attention Bank

Item	Value
Needs training	No
Needs ref image	No
Extra compute	Low medium
Identity stability	0.80
Best for	Zero shot comics
Limitation	Far shots drift

3 Metrics and Results

We used two metrics: CLIP score for image and prompt alignment. Dialogue perplexity (GPT-2) for dialogue text.

Story perplexity: 52.69924030014943. Panels 1–4 are identity reference frames, so the main results start from Panel 5.

Panel ID	CLIP Score	Dialogue PPL
Panel 5	0.333602	3379.894791
Panel 6	0.286558	405.132932
Panel 9	0.280402	4110.204354
Panel 10	0.380081	305.397838
Panel 11	0.328300	919.613643
Panel 12	0.353496	1531.153958
Panel 13	0.318106	3063.134519
Panel 14	0.331733	333.158882
Story PPL	52.69924030014943	

Table 4: Results for selected panels

4 Prompt and Image Examples

We show a few example panels (full prompts).

Panel	Image	CLIP	Prompt (full)
10		0.380081	single comic panel, centered, clean ink lineart, simple cel shading, vibrant colors, high quality, detailed faces, no text, Bathroom shot, Household room, Washing hands, Soapy water, Boy A on the left: curly black hair, blue denim overalls, shy smile; Boy B on the right: straight brown hair, red hoodie, playful grin, exactly two kids
12		0.353496	single comic panel, centered, clean ink lineart, simple cel shading, vibrant colors, high quality, detailed faces, no text, Playground shot, Outdoor play area, Climbing structures, Supporting each other, Boy A on the left: curly black hair, blue denim overalls, shy smile; Boy B on the right: straight brown hair, red hoodie, playful grin
14		0.331733	single comic panel, centered, clean ink lineart, simple cel shading, vibrant colors, high quality, detailed faces, no text, Porch shot, Outside entrance, Relaxing, Sharing fan, Boy A on the left: curly black hair, blue denim overalls, shy smile; Boy B on the right: straight brown hair, red hoodie, playful grin, exactly two kids

Table 5: Sample panels with full prompts and CLIP scores

5 Analysis

CLIP scores are mostly mid range, showing acceptable prompt alignment. Panel 10 has the highest score (0.380081), while Panels 6 and 9 are lower, so some small details are missed.

Dialogue perplexity is high in many panels, so dialogue quality is not stable by this metric.

The Attention Bank method improves identity consistency without training or reference images. Using id_length = 4 builds identity memory from the first four panels. sa32 = 0.85 helps keep main face features and sa64 = 0.75 keeps details while allowing pose and background changes. Drift can still appear in far shots or very small faces, and hands or small accessories may change.

6 Conclusion

The project is not production-ready because it requires substantial GPU compute for Qwen2.5-7B and SDXL.